इलेक्ट्रॉनिकी एवं सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY

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DSCI
PROMOTING DATA PROTECTION
A nasscom Initiative

CYBER SECURITY INNOVATION CHALLENGE 1.0

DRIVING SECTOR-RELEVANT & FUTURE-READY CYBERSECURITY SOLUTIONS

Cluster : Cryptography

Problem Statement title : Privacy Preserving KYC Verification System

Description:

Design a privacy-preserving KYC verification system leveraging applied privacy-enhancing technologies (PETs), such as zero-knowledge proofs (ZKPs), that allows users to prove identity attributes (e.g., 'over 18', 'valid Aadhaar') without revealing sensitive personal identifiable information (PII).

Exact Deliverables :

- Prototype of a privacy-preserving KYC system with user-friendly verification interfaces.

- Integration of selective-disclosure or zero-knowledge proof mechanisms.

- Performance benchmarking report comparing latency and usability against traditional e-KYC.

- Demonstration of compliance with India’s DPDP Act (2023) requirements.

Milestones, Evolution Parameters:

- Phase 1: Basic prototype with ZKP-based “age over 18” verification.

- Phase 2: Expand to multi-attribute verification (e.g., address, Aadhaar validity).

- Phase 3: Integration with mock financial/telecom onboarding systems.

- KPIs: Verification latency vs. e-KYC, PII exposure reduction score, and user adoption experience (measured via mock testing).

Additional Information:

- Students should benchmark their systems against both Aadhaar e-KYC APIs and privacy-preserving identity protocols (e.g., Hyperledger Indy, W3C Verifiable Credentials).

- Focus on building usable, intuitive interfaces—privacy-preserving systems often fail due to poor UX.

- Systems must balance compliance with practicality; lightweight implementations that can scale are preferred.